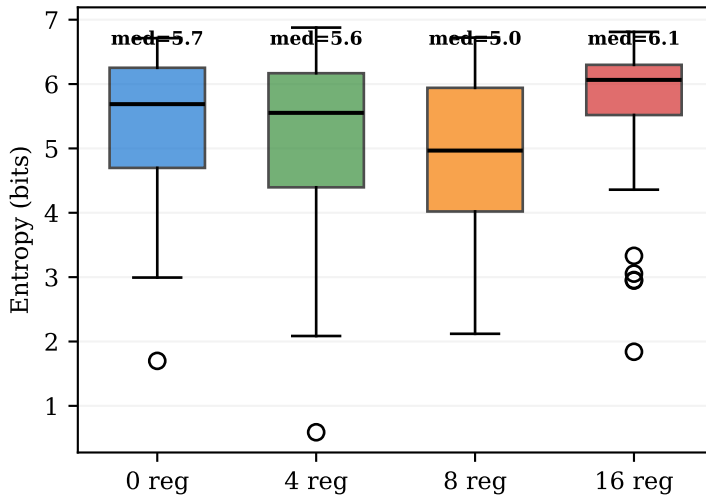
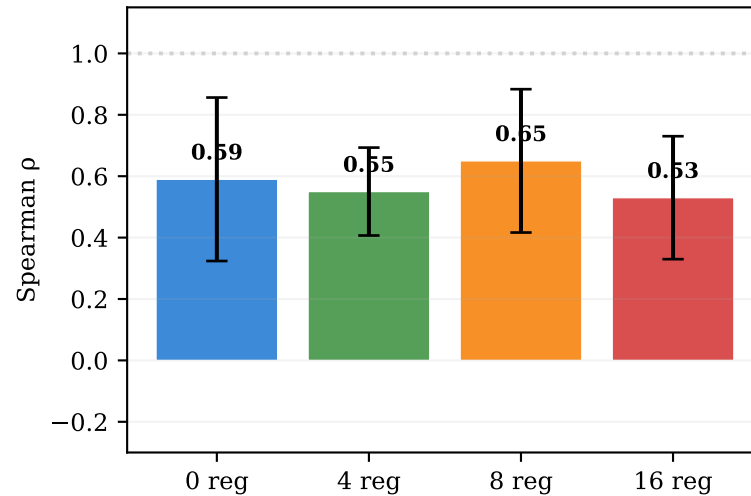


Quantitative Attention Quality — Register Effect

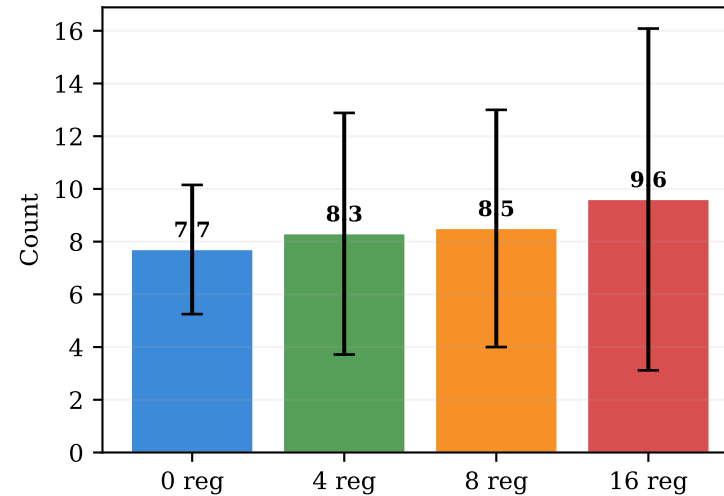
(a) Character Attention Entropy
lower = more focused



(b) Spatial Ordering Quality
higher = better L→R alignment



(c) Aligned Attention Heads
($\rho > 0.5$ per head)



(d) Token Norm Distribution
register vs patch tokens

